

Apuntes: Fulton's Algebraic Curves

(detesto el Fulton)

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3.1 Comentarios

El objetivo de estos apuntes es lograr armar una suerte de acompañamiento para el Algebraic Curves de Fulton. Mientras más lo voy leyendo, más me parece que es sumamente complicado seguirlo, dado lo compacto que es y la cantidad de detalles que se suele saltar. Por lo tanto, esto nace por la incompetencia que siento cada vez que lo retomo.

En particular, quiero que este apunte se centre en 4 cosas principalmente:

1. Definiciones;
2. Resultados importantes;
3. Comentarios respecto a las demostraciones que se entregan;
4. Comentarios respecto a la posible intuición que se puede tener en un determinado momento / Explicaciones de cómo pienso **yo** la materia.

Chapter 4

Projective Varieties

4.1 Projective Space

Definition 4.1. Let k be any field. We define the *projective n space* over k , written as $\mathbb{P}^n(k)$ (or simply $\mathbb{P}^n$), with the set of equivalence classes of points in $\mathbb{A}^{n+1} \setminus \{(0, \dots, 0)\}$, where two points $(x), (y) \in \mathbb{A}^{n+1} \setminus \{(0, \dots, 0)\}$ are $(x) \sim (y)$ if and only if there is a nonzero $\lambda \in k$ such that $x_i = \lambda y_i$ for all $i = 1, \dots, n+1$.

Remark. It's easy to see that we are relating points if they are on the same (unique) line that passes through the origin on $\mathbb{A}^{n+1}$. Then, points in $\mathbb{P}^n$ are lines passing through the origin in $\mathbb{A}^{n+1}$.

Definition 4.2. We let $U_i = \{[x_1 : \dots : x_{n+1}] \in \mathbb{P}^n \mid x_i \neq 0\}$. Each $P \in U_i$ can be written uniquely in the form

$$P = [x_1 : \dots : x_i : \dots : x_{n+1}].$$

Then, we define $\varphi_i: \mathbb{A}^n \rightarrow U_i$ by $\varphi_i(a_1, \dots, a_n) = [a_1 : \dots : a_i : \dots : a_n]$, which defines a one-to-one correspondence between the points of $\mathbb{A}^n$ and the points of U_i .

Remark. Note that $\mathbb{P}^n = \bigcup_{i=1}^n U_i$, so $\mathbb{P}^n$ is covered by $n+1$ sets each of which look just like affine n -space.

Definition 4.3. Let

$$H_\infty := \mathbb{P}^n \setminus U_{n+1} = \{[x_1 : \dots : x_{n+1}] \mid x_{n+1} = 0\};$$

H_∞ is often called the *hyperplane at infinity*.

Remark. The correspondence $[x_1 : \dots : x_n : 0] \leftrightarrow [x_1 : \dots : x_n]$ shows that H_∞ may be identified with $\mathbb{P}^{n+1}$. Thus, $\mathbb{P}^n = U_{n+1} \cup H_\infty$ is the union of an affine n -space and a set that gives all directions in affine n -space.

4.2 Projective Algebraic Sets

Definition 4.4. A point $P \in \mathbb{P}^n$ is said to be a *zero* of a polynomial $F \in k[X_1, \dots, X_{n+1}]$ if

$$F(x_1, \dots, x_{n+1}) = 0$$

for every choice of homogeneous coordinates $(x_1, \dots, x_{n+1})$ for P ; we then write $F(P) = 0$.

Definition 4.5. For any set S of polynomials in $k[X_1, \dots, X_{n+1}]$, we let

$$V(S) = \{P \in \mathbb{P}^n \mid P \text{ is a zero of each } F \in S\}.$$

Remark. If I is the ideal generated by S , $V(I) = V(S)$.

Definition 4.6. If $I = (F^{(1)}, \dots, F^{(r)})$, where $F^{(i)} = \sum F_j^{(i)}$, $F_j^{(i)}$ a form of degree j , then $V(I) = V(\{F_j^{(i)}\})$, so $V(S) = V(\{F_j^{(i)}\})$ is the set of zeros of a finite number of forms. Such a set is called an algebraic set in $\mathbb{P}^n$, or a *projective algebraic space*.

Definition 4.7. For any set $X \subset \mathbb{P}^n$, we let

$$I(X) = \{F \in k[X_1, \dots, X_{n+1}] \mid \text{every } P \in X \text{ is a zero of } F\}.$$

The ideal $I(X)$ is called the *ideal* of X .

Definition 4.8. An ideal $I \subset k[X_1, \dots, X_{n+1}]$ is called *homogeneous* if for every $F = \sum_{i=1}^m F_i \in I$, F_i a form of degree i , we have also $F_i \in I$.

Remark. For any set $X \subset \mathbb{P}^n$, $I(X)$ is a homogeneous ideal.

Proposition 1. An ideal $I \subset k[X_1, \dots, X_{n+1}]$ is homogeneous if and only if it is generated by a (finite) set of forms.

4.3 Affine and Projective Varieties

4.4 Multiprojective Space

Chapter 5

Projective Plane Curves

5.1 Definitions

5.2 Linear Systems of curves

5.3 Bezout's Theorem

5.4 Multiple Points

5.5 Max Noether's Fundamental Theorem

Corollary 1. If either

- (1) F and G meet in $\deg(F)\deg(G)$ distinct points, and H passes through these points, or
 - (2) All the points of $F \cap G$ are simple points of F , and $H \bullet F \geq G \bullet F$,
- then there is a curve B such that $B \bullet F = H \bullet F - G \bullet F$.

5.6 Applications of Noether's Theorem

Proposition 1. Let C be an irreducible cubic, C' , C'' cubics. Suppose $C' \bullet C = \sum_{i=1}^9 P_i$, where the P_i are simple (not necessarily distinct) points on C , and suppose $C'' \bullet C = \sum_{i=1}^8 P_i + Q$. Then $Q = P_9$.

Remark. In the proof of the proposition (number 3 on Fulton's book), it's stated that when one gets $LC'' \bullet C = C' \bullet C + Q + R + S$, you can then say that there is a line L' such that $L' \bullet C = Q + R + S$. Effectively, this comes from the corollary, part (2), presented in section 5.5. Indeed, if we take $F := C$, $G := C'$, and $LC'' := H$, we can first see that the points on $F \cap G = C \cap C'$ are exactly the P_i 's, all simple points of C . Also, because of the equality given at the beginning

of the note, it's easy to see that

$$H \bullet F = LC'' \bullet C \geq C' \bullet C = G \bullet F.$$

Then, because of the corollary, there is a curve B such that

$$B \bullet C = LC'' \bullet C - C' \bullet C = Q + R + S.$$

This B must be a line, because $\sum_P I(P, C \cap B) = 3 = 3 \cdot \deg(B)$; i.e., $\deg(B) = 1$. So $B =: L'$ is the line that we were looking for.

Chapter 6

Varieties, Morphisms, and Rational Maps

6.1 The Zariski Topology